# Alcohol Hangover Syndrome and Driving-Related Cognition: A Dual-Task and Modelling Approach

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16th June 2025

## Acknowledgements

I would like to begin by thanking my supervisors, Dr. Matthew Albrecht and Dr. Razi Hasan, for their continued guidance and support through the course of this thesis. I am incredibly grateful for their support when my initial project fell through and I had to change research directions midway. Their patience and belief in me helped me stay motivated during times when I was struggling to stay on track, and without their mentorship, this thesis would not have been possible.

I would also like to acknowledge the contributions of researchers from the Western Australia Centre for Road Safety Research who were involved in the design and data collection for this study. Their initial work helped me formulate the research questions that shaped this thesis.

Finally, I would like to thank the participants who participated in this study and their key contributions that made this research possible.

# Declaration

I declare the work presented in this dissertation is my own, unless stated otherwise. This work has not been submitted previously for any degree or qualification at this institution or any other.

The research presented here is based on data that was originally collected by Monique du Toit, a current Ph.D student at the University of Western Australia and Tex Garstone, a former UWA School of Psychological Sciences student as part of their honours dissertation. The participants recruited were undergraduates from the UWA School of Psychological Sciences via the SONIA online portal.

Dr. Matt Albrecht wrote the code for extracting the DDM parameters based on a script from the open source implementations of Stan DDM by Marco Wirthlin. All subsequent data analysis, interpretation, and writing were carried out independently by me.

Signed: Yatin Azad
Dated: 16.06.2025

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# List of Abbreviations

| | |
|---|---|
| ADHD | Attention Deficit Hyperactivity Disorder |
| AHS | Alcohol Hangover Scale |
| AHSS | Alcohol Hangover Symptom Severity Scale |
| BAC | Blood Alcohol Concentration |
| DDM | Drift Diffusion Model |
| EZ DDM | Easy DDM |
| HDDM | Hierarchical Drift Diffusion Model |
| LMM | Linear Mixed Effects Model |
| MAD | Mean Absolute Deviation |
| MCMC | Markov Chain Monte Carlo |
| RT | Reaction Time |
| SRTT | Simple Reaction Time Task |
| SIHS | Single Item Hangover Scale |
| WHO | World Health Organisation |
| | |

# Abstract

Subtle psychomotor and cognitive changes related to alcohol hangover compared to intoxication require a radically different research approach due to potentially ignored real-world driving risks. Previous studies had reported mixed results employing computational models to dissect the cognitive processes. We aimed to investigate whether hangover affects decision-making, psychomotor and temporal estimation processes. We hypothesised that subjects would show reduced psychomotor speed and higher variability in anticipatory timing while being hungover, reflecting a deterioration of these processes. The present study involved twenty subjects with mild hangover (AHSS-*m*-20.4, *sd*-12.5) who underwent a randomised crossover semi-naturalistic study using validated questionnaires. A novel designed hierarchical drift diffusion model and an anticipatory timing task were used to investigate the effects of alcohol-hangover on the underlying cognitive components of response processing and mean absolute deviation (MAD) across four different conditions.

This study showed that mild hangovers had a limited impact on drift rate (*v)* ($p$ = 0.61), which was hard to interpret given crossover interactions of state and sessions ($p$ = 0.48). On the other hand, inter-trial variability parameters for drift rate (*sv*) ($p$ = .049) and non-decision time (*st0*) ($p$ = .044) were significantly affected by the hangover state ($p$ = .015). While the analysis also uncovered main effects for both state($p$ = .042) and session ($p$ = .026) in predicting reaction time task performance, these effects were not clearly differentiable due to the crossover interaction. Contrary to our hypothesis, no significant main effect of state emerged for temporal estimation variability(MAD) in any of the four task conditions. Exploratory covariate analysis showed that observed effects of hangover severity and state in the decision-making process are not explained by the self-reported measures of sleep and fatigue**.**

The study found some basic changes in psychomotor speed and slower evidence accumulation, but the main effects could have been vitiated by session and state, in addition to a mild hangover state. Further research incorporating functional neuroimaging and real-time electro-physiological monitoring across a broader range of hangover severities is needed to uncover the true underlying impact of alcohol hangover on driving-relevant cognition.

# Introduction

Alcohol hangover syndrome can be defined by a set of unpleasant physical and psychological symptoms felt after heavy alcohol intake in a single episode when blood alcohol concentration (BAC) has almost returned to zero. Hangover is known to affect cognitive domain, job productivity, and transportation safety (Verster et al., 2020; Gunn et al., 2018). While excessive alcohol intake contributes to a significant portion of road accident-related fatalities with regional variations (WHO, 2018), hangovers are an often underestimated and insidious public health menace. A combination of cognitive and physical symptoms, like headache, fatigue, impaired concentration, and slowed reaction times, is known to persist for hours in this state (Verster et al., 2020; Gunn et al., 2018). Such individuals are known to exhibit limitations in their ability to perform complex tasks like driving.

Individuals with hangover symptoms can have this potential misbelief that they are fit to drive, with evidence suggesting otherwise (Verster et al., 2020). Despite a lack of precise data for hangover driving, many studies suggest it is a common behaviour with potentially severe consequences (Verster et al., 2010; Robbins et al., 2019). Presently, a hangover state cannot be detected by any readily available roadside test, unlike alcohol intoxication.

Symptoms of alcohol hangover could result from mechanisms like inflammation, gastrointestinal irritation, acetaldehyde exposure, dehydration, hormonal imbalance, and mini-withdrawal, while severity may be affected by age and type of beverage, including congeners (Leclercq et al., 2019; Penning et al., 2010). Elevation of pro-inflammatory markers like interleukin-6 (IL-6), cytokines, tumour necrosis factor-alpha (TNF-α), and C-reactive protein (CRP) may support their significant role in leading to cognitive dysfunction and subjective distress during hangovers (Ayre et al., 2021).

Studies have found that both core and higher-order cognitive processes are impaired during alcohol hangover (Scholey et al., 2019; Ayre et al., 2021). Initial research into alcohol hangovers was mostly based on hangover scales and questionnaires, scrutinising general cognitive and psychomotor deficits (Gunn et al., 2018; Verster et al., 2020). Many validated scales of assessment include the Acute

Hangover Scale (AHS) (Rohsenow et al., 2007), the Hangover Symptoms Scale (HSS) (Slutske et al., 2003), the Single Item Hangover Severity (SIHS) (Verster et al., 2020) and the Alcohol Hangover Severity Scale (AHSS) (Penning et al., 2013). These scales helped to decipher individual susceptibility to hangovers by combining the scoring of individual frequency of symptoms. These findings can give us initial insights, but a deeper understanding of driving-related hangover derangements requires a more focused and targeted approach.

Driving simulators and on-road tests have been used previously for hangover research and detected impairments in several key driving abilities. Increased lateral deviation, increased variability in speed control, slower reaction times, metacognitive deficits, and impaired insights were observed in driving in a hungover state (Verster et al., 2014; Verster & Roth, 2014; Alford et al., 2020).

Cognitive impairments in hangover are difficult to recognise and require more rigorous analytical tools and modelling approaches. Recently, the hierarchical drift diffusion model (HDDM) has been used to assess decision-making in various areas of psychological and neuroscientific research. HDDM is a quantitative model that decomposes choice decision-making into distinct latent psychological processes, including the rate of evidence accumulation (drift rate, $v$), response caution (boundary separation, $a$), and non-decision time ($t_0$) (Ratcliff & McKoon, 2008). Only a few previous studies have applied DDM explicitly to hangover states (Grange et al, 2016; Stock et al, 2017) as opposed to intoxication, representing a significant gap in the literature. Many variants of DDM have been used previously with mixed results in hangover states (Grange et al, 2016; Ratcliff et al., 2004; Wagenmaers et al, 2007). The HDDM in this study was implemented using a custom model script.

Assessment of mean simple reaction time, although informative, may not be suited to assess a complex real-world skill like driving ( Grange et al, 2016; Devenney et al., 2019). As a measure of central tendency, mean reaction time only provides an overall summary of response speed and may fail to capture inter-trial variability and state-dependent fluctuations in performance, which is more valuable to high-stakes environments like driving.  Driving often requires not just rapid responses, but consistent and accurate timing of actions. Therefore, this study also employed a unique anticipatory timing task, which requires participants to predict the arrival of a moving object across four different states. We wanted to evaluate performance consistency by

quantifying response variability as the mean absolute deviation (MAD). This measure has been used previously but not in hangover research (Cicchini GM et al, 2012; Jazayeri M et al, 2010). Measuring the consistency of responses (MAD) rather than just the average accuracy provides a more sensitive index of performance stability, which may be particularly susceptible to the effects of hangover and, in turn, affect driving ability.

The primary aim of the current study was to provide a detailed analysis of the effects of alcohol hangover on driving-related cognition using these advanced methods and a semi-naturalistic design. The participants completed various questionnaires for basic demographics, sleep diary, hangover scale, and severity, which could impact decision-making. Objectively, the study sought to:

- Assess baseline psychomotor speed using a simple reaction time task (SRTT)
- Predict temporal estimation variability using a visuospatial anticipatory motor timing task.
- Uncover the key changes in response processing between hungover and sober states using HDDM.
- Investigate the impact of subjective hangover severity and sleep measures on objective performance measures in the hungover state.

It was hypothesised that participants impaired by the effects of alcohol hangover would show:

- Slower psychomotor speed, as shown by longer reaction times in the SRTT.
- Reduced drift rate (*v*) and increased variability in non-decision time and drift rate (*sv*, *st0*), suggesting noisier processing and slower evidence accumulation.
- Greater variability in temporal estimation (higher MAD) in the anticipatory timing task. More demanding task conditions(occlude, sped up) would correlate with increased variability.
- Subjective ratings of fatigue, sleep and hangover severity would be significant predictors of task performance, even after accounting for the main effect of hungover state.

With this study, we wanted to explore the gaps in the literature by examining hangover effects on driving-relevant cognitive processes by incorporating subjective self-

assessments and objective performance measures in a novel task paradigm combined with tailored computational modelling.

# Methods

## Participants

A total of 20 undergraduate students were recruited from the University of Western Australia's School of Psychology participant pool and through word-of-mouth. Inclusion criteria required that all hold a valid driver's license and be over 18 years of age. Participants were required to arrange transportation to and from the laboratory on the hangover testing day that did not require them to drive their own vehicle as a precaution against possible hangover impairment or BAC still above 0.05. All participants provided written informed consent prior to their involvement in the study and received course credits for their participation. The research received ethical approval from the University of Western Australia's Human Research Ethics Board (2023/ET000004).

## Study Design

The study employed a within-subjects, randomised, crossover design. Each participant attended two testing sessions at the laboratory. For the Hangover session, participants were instructed not to change their drinking behaviour for the study and to record their drink intake. For the Sober session, participants were asked to abstain from alcohol for at least 24 hours prior to testing. The order of these conditions was counterbalanced across participants to mitigate potential order effects, like practice or fatigue, confounding the main state effect.

Upon arrival at the laboratory for each session, participants' Blood Alcohol Concentration (BAC) was measured using a breathalyser to confirm a reading of 0.00%. This ensured that any observed effects were due to the residual effects of hangover and not the acute effects of alcohol. Following the BAC test, participants completed the set of questionnaires. They were then seated at a computer and completed the cognitive and motor timing tasks in a fixed order, beginning with the SRTT, followed by the anticipatory timing task. Each testing session lasted approximately 60 minutes, including setup and scheduled breaks. At the conclusion of the second session, participants were fully debriefed.

## Materials and Measures

All tasks were administered on a standard HP computer on a L2245wg 22-inch monitor, with 1680x1050 pixels and a refresh rate of 60 Hz. A range of validated questionnaires and cognitive tasks were utilised. All of the cognitive testing occurred on the Inquisit 6 (2020) software platform developed by Millisecond.

### *Questionnaires*

A battery of questionnaires was administered to collect demographic data and assess participants' subjective state during each session.

- **Demographics and Background Questionnaire.** This survey collected baseline information on participant age, gender, ethnicity, and driving history, including years of driving experience. It also screened for pre-existing eye conditions or attention disorders (e.g., ADHD) that could potentially affect performance on the tasks.
- **Alcohol Consumption.** Participants reported the types and number of standard drinks consumed the night prior to the session.
- **Alcohol Hangover Severity Scale (AHSS).** Participants self-reported the presence and severity of a range of common hangover symptoms. This is a 15-item visual analogue scale adapted from Penning et al. (2012b) with values ranging from 0 to 100, with 0 indicating the absence of that particular symptom. Most studies reported a Cronbach's $a > 0.8$, indicating high reliability of the AHSS as a hangover severity measure (Penning et al., 2012b, 2013). The "Fatigue" subscale from this instrument was used as a specific covariate in the analysis.
- **Single Item Hangover Severity (SIHS).** Participants rated their overall hangover severity on a visual analogue scale from 0 ("no hangover symptoms") to 100 ("extreme hangover symptoms"). While not a scale in the psychometric sense, it is widely used for its simplicity and strong correlations with other multi-item hangover scales (Verster et al., 2010; Penning et al., 2013). It is also highly sensitive to state-based measures, with Verster et al. (2020) suggesting it may even be superior to AHSS in assessing overall hangover severity.
- **Drunkenness Rating:** Participants provided a retrospective rating of their

peak level of intoxication on a scale of 0-100.

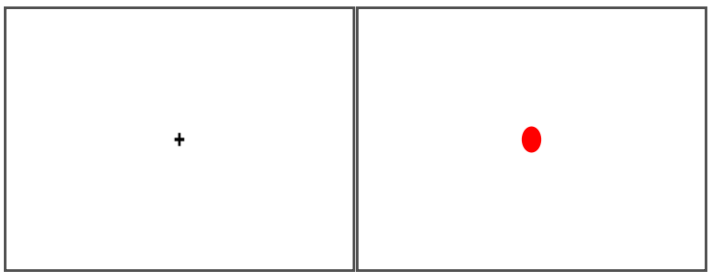
- **Sleep Questionnaires:** An abbreviated 8-item Consensus Sleep Diary was used to assess subjective sleep quality and quantify sleep duration for the night immediately preceding each testing session (Carney et al, 2012). The complete CSD has excellent test-retest reliability ($r > 0.85$) and high convergence with other actigraphy-based sleep measures ($r \sim 0.68 - 0.76$). Even in its abbreviated form, the diary retains its psychometric properties, with Cronbach's $a > 0.80$ in clinical and student populations (Maich et al., 2018). Ratings reported in this questionnaire were used to compute Sleep Efficiency as the ratio of total sleep duration to the total time spent in bed.

***Computerised Cognitive Testing***

**Simple Reaction Time Task.** A standard reaction time task from the Millisecond library (Simple Reaction Time Task - Millisecond, 2024) was administered to assess baseline psychomotor speed as presented in Figure 1. The participants were presented with a fixation cross, followed by a visual target stimulus (a red circle) in variable time intervals randomly selected from 2000 ms to 8000 ms. The participants were instructed to press the spacebar as soon as the target stimulus appeared on the screen. Latency between the target stimulus and the key press was recorded for each trial. Data from this task formed the basis for drift diffusion modelling analysis.

**Figure 1**

*Simple Reaction Time Task*



*Note*. A fixation cross was followed by a red circle, which the participants responded to.

**Anticipatory Motor Timing Task.** A custom-designed Anticipatory Timing Task was administered to assess temporal estimation variability. In all conditions, the core task required participants to watch a black disk appear at the top of the screen and fall towards a blue target bar. Their objective was to press the SPACEBAR at the exact moment the disk would first touch the target, anticipating the impact rather than

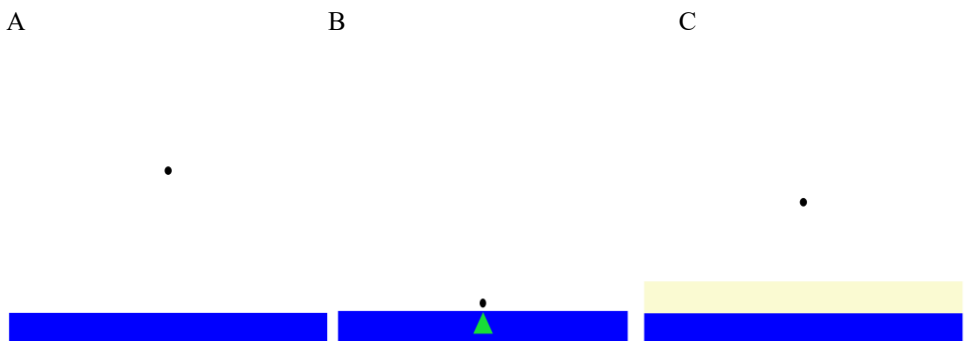
reacting to it. Participants would receive feedback on their delay during practice trials presented as a milliseconds deviation from the timing of the press relative to when the disk touched the target. The task comprised four conditions, presented in a sequential order as shown in Figure 2(A-C):

1. Baseline Condition**:** The disk fell at a constant speed, and its full trajectory was visible. The time taken to reach the target was 2000 ms. This measured baseline motor timing estimation (Figure 2A).
2. Distracted Condition**:** The task was identical to baseline, but a green distractor symbol appeared 1600 ms before the disk reached the target. Participants were instructed to ignore the symbol. This assessed motor timing estimation under visual distraction (Figure 2B).
3. Occluded Condition**:** The disk fell at a constant speed but disappeared behind a yellow "occlusion zone", 1600 ms before reaching the target. Participants had to estimate the moment of impact without a reference visual cue. This measured timing based on internal models of speed and distance approximations (Figure 2C).
4. Speed Variation Condition**:** The speed at which the disk fell varied between 2 speeds (50% or 75% quicker than baseline) from trial to trial, requiring participants to constantly re-evaluate and adjust their temporal predictions (Figure 2A).

For the four temporal estimation task conditions, response time variability was quantified as the Mean Absolute Deviation (MAD) in temporal estimation error. Here, the temporal estimation error was calculated as a millisecond deviation from the timing of the press relative to when the disk touched the target.

**Figure 2**

*The visual conditions of the Anticipatory Timing Task*



*Note.* (A) Control and Speed condition, which had the disk falling at variable speeds to the target, (B) the Distracted condition, with an irrelevant green symbol appearing near the target, and (C) the Occluded condition, where the disk's final trajectory is hidden.

### *Drift Diffusion Modelling (DDM)*

The DDM is a sequential sampling model that enables the simultaneous estimation of group-level and individual-level parameters, providing more robust estimates for each participant. It is typically used to decompose performance in reaction time tasks into distinct psychological parameters. The current study utilised an HDDM model, adapted from the single-choice reaction time DDM model by Ratcliff within a hierarchical framework (Ratcliff et al., 2002; 2006). A hierarchical approach permits simultaneous estimation of group-level and individual participant parameters.

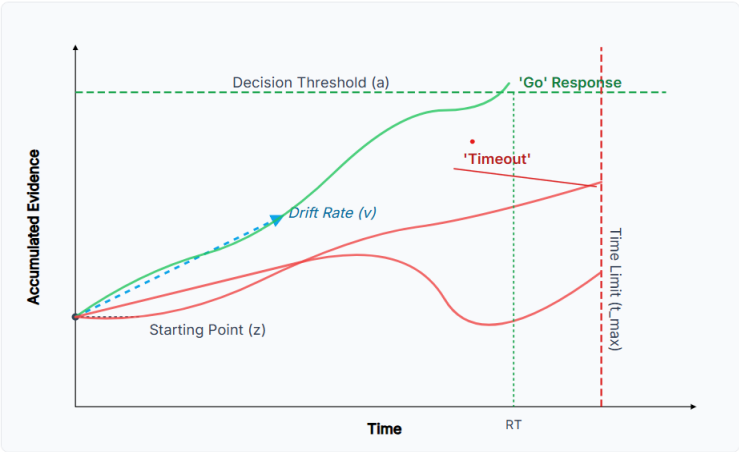
The HDDM was implemented using the StanDDM R package (Wirthlin et al., 2025), using a custom model script. The model estimated the following parameters for each participant for each session:

- Drift Rate **(*v*):** To assess the mean rate of evidence accumulation.
- Non-Decision Time **($t_0$):** To evaluate the time used for perceptual encoding and motor response execution, processes that were not directly related to the decision.
- Inter-Trial Variability of Drift Rate **(*sv*):** To account for trial-to-trial fluctuations in the rate of evidence accumulation.
- Inter-Trial Variability of Non-Decision Time **($st_0$):** Accounts for trial-to-trial fluctuations in non-decision processes.

The Boundary Separation **($a$ = 1.0)** and Starting point ($z$ = 0.5*)* were fixed to enable the DDM to fit a single-choice reaction time test. These parameters are visualised in Figure 3.

**Figure 3.**

*The Single-choice drift diffusion model (DDM).*




*Note.* The vertical axis represents the accumulation of sensory information. The x-axis represents the time since stimulus presentation, and RT indicates the time at which a response was recorded. If 10000ms had elapsed since stimulus presentation, a no response was recorded for that particular trial.

**Data Preparation for DDM.** Reaction times (RTs) from the reaction time task were used as input for the DDM. Trials with RTs longer than 1.2 seconds were excluded from the analysis to remove overly slow responses. Model parameters were estimated using Bayesian inference via Markov Chain Monte Carlo (MCMC) sampling, as implemented in Stan (Stan. R package version 2.32.7). For each model fit, four MCMC chains were run. Each chain consisted of 800 iterations, and the first 400 iterations were discarded as warmup. Thus, 1600 samples were taken from the posterior distribution for each parameter. A fixed random seed was used for reproducibility.

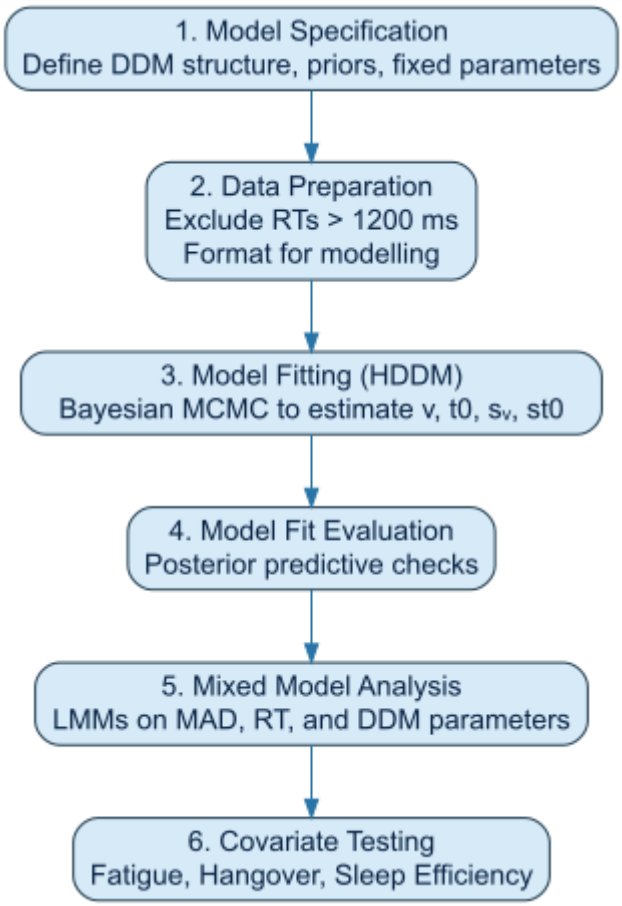
Consistent with common practice in HDDM literature, weakly informative priors were fixed for the group-level hyperparameters. The prior for the overall group mean drift rate "$v$" was described by a normal distribution with mean = 5 and SD = 3 [i.e., $v$ ~Normal(5,3)] and for group mean non-decision time, it was $t_0$ ~ Normal (0.2,0.2). Subject-level parameters were subsequently drawn from these group-level distributions. Model convergence was assessed via visual inspection of the trace plots to

ensure proper mixing and stationarity. R-hat ($\hat{R}$) (typically < 1.1) values were also verified for all parameters. Individual participant parameter estimates for ($v$, $t_0$, $sv$, $st_0$) were obtained by taking the mean of their respective posterior distributions derived from the MCMC samples.

**Model Convergence Diagnostics**. The adequacy of the DDM was confirmed through additional posterior predictive checks, which involved comparing the simulated reaction time distributions from the fitted model to the observed data. These were performed at both the individual participant level and the aggregated group level. The estimated posterior means for each parameter for each participant were then compared across different states and sessions using mixed regression modelling. An overview of key steps in the modelling process is presented in a flow chart depicted in Figure 4**.**

**Figure 4**

***The six-step analysis pipeline***




**Statistical Analysis**

To account for the repeated measures design of the study, we applied Linear Mixed-Effects Models (LMMs) using the lmer function from the lme4 package (Bates

et al., 2015) in R (Version 4.4.3, 2025). This framework allowed us to model both fixed effects and random effects simultaneously. The dependent variables for the model included mean reaction time in the SRTT, DDM parameters from the fitted SRTT data, and motor timing variability (MAD), across the various motor timing task conditions (as described in Section X).

For each dependent variable, the primary model was specified with State (Sober, Hungover), Session (1, 2), and their interaction (State * Session) as fixed effects. The Session factor was included to help control for any potential order effects, such as practice or fatigue, that could occur between the two testing occasions. Including the interaction term helped examine whether the effect of State was impacted by task familiarity across the two sessions. A random intercept for each participant was included to account for the non-independence of observations, since some participants can be naturally faster/slower than others. To explore the predictive utility of other covariates, the base LMM for each MAD variable was compared against more complex models that additionally included Hangover Severity (Single Item Hangover Scale), Fatigue, or Sleep Efficiency. The Satterthwaite method was used to estimate degrees of freedom and calculate *p*-values for the fixed effects (Kuznetsova et al., 2017). The residuals of each final model were assessed for normality via visual inspection of Q-Q plots to ensure the validity of the statistical assumptions. A significance level of 0.05 was used for all inferential tests.

# Results

## Participant Characteristics and Baseline Measures

Table 1 presents demographics for the sample with hangover severity and sleep measures separated by state.

**Table 1**

*Participant Demographics and Key Variables (N = 20)*

| Variable | Sober<br>*M (SD)* | Hungover<br>*M (SD)* | Range<br>(Sober / Hangover) |
|---|---|---|---|
| Gender (% female) | 14 (70%) | — | — |
| Gender (% male) | 5 (25%) | — | — |
| Gender (% other) | 1 (5%) | — | — |
| Ethnicity (% Caucasian) | 10 (50%) | — | — |
| Ethnicity (% Asian) | 4 (20%) | — | — |
| Ethnicity (% Mixed/Other) | 6 (30%) | — | — |
| Eye conditions (%) | 6 (30%) | — | — |
| Age (years) | 22.6 (8.1) | — | 19–56 |
| Years driving | 5.0 (8.2) | — | 1–38 |
| Standard Drinks Consumed | — | 7.35 ( 4.89 ) | 0 - 15 |
| Drunkenness rating | — | 38.8 (21.0) | 1–86 |
| Single-Item Hangover Severity (SIHS) | 2.0 (4.2) | 14.2 (13.9) | 0–10 / 0–55 |
| Sleep duration (hours) | 7.4 (1.3) | 6.5 (1.4) | 5.0–9.3 / 3.4–9.2 |
| Sleep efficiency | 0.89 (0.06) | 0.84 (0.10) | 0.80–1.00 / 0.52–0.96 |

| | | | |
|---|---|---|---|
| Alcohol Hangover Severity Scale(AHSS) | 9.7 (8.8) | 20.4 (12.5) | 0.2–28.2 / 2.5–40.9 |
| Fatigue | 23.4 (20.4) | 47.6 (22.4) | 1.0–60.0 / 10.0–84.0 |

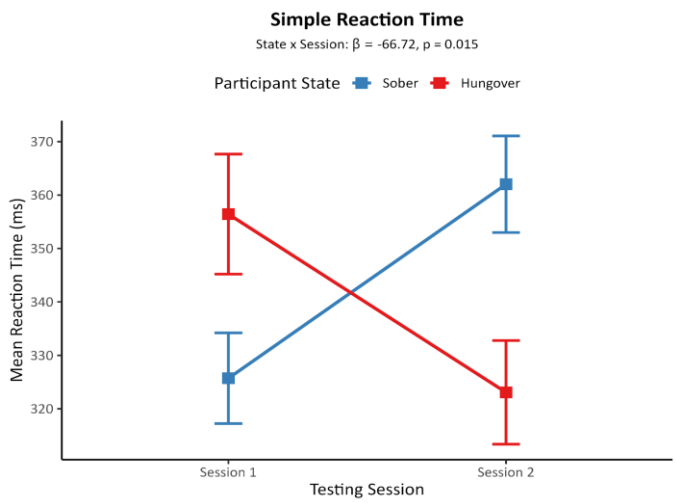
As expected, hungover participants reported higher hangover severity on the Single Item Scale ($t$ (19) = 3.49, $p$ = 0.003) and greater fatigue ($t$ (19) = 3.85, $p$ = 0.001), along with reductions in sleep duration ($t$ (19) = -2.20, $p$=0.042) and efficiency ($t$ (19) = -1.97, $p$=0.046). One participant was excluded from the analysis for failing to respond effectively to certain conditions of the anticipatory timing task.

**Simple Reaction and Psychomotor Performance**

We first assessed basic psychomotor functioning using the SRTT. A linear mixed-effects model pointed towards a significant state by session interaction ($\beta$ = -66.72, $t$(17.31) = -2.69, $p$ = .015), indicating that hangover effects on reaction time were dependent on the testing session, as seen in Table 2. Figure 5 presents this crossover interaction, showing that sober participants' reaction times increased from session 1 to session 2, whereas hungover participants' reaction times decreased across the two sessions. While the analysis also uncovered main effects for both state ($\beta$ = 30.72, $t$(28.58) = 2.13, $p$ = .042) and session ($\beta$ = 34.21, $t$(29.13) = 2.34, $p$ = .026), these effects are not clearly differentiable due to the presence of the crossover interaction.

**Figure 5**

*Simple reaction time by state and session.*



*Note.* Points represent the mean reaction time for each state for all N=19 participants within each session. Error bars depict the standard error of the mean.

**Table 2**

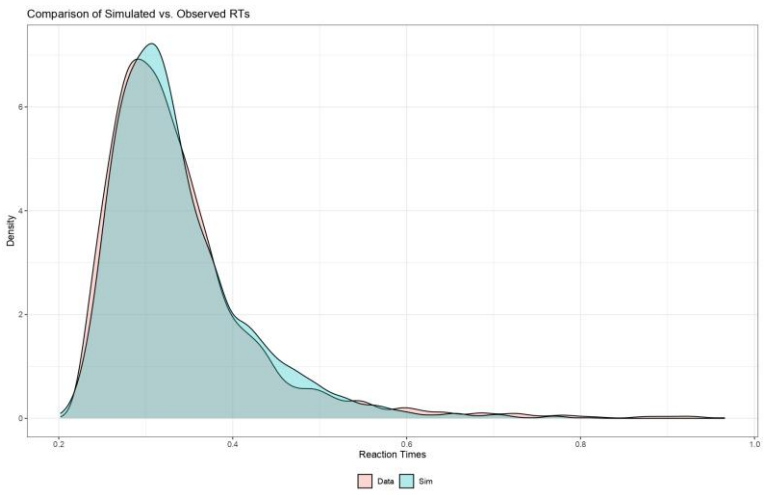
*Fixed Effect Estimates for Simple Reaction Time.*

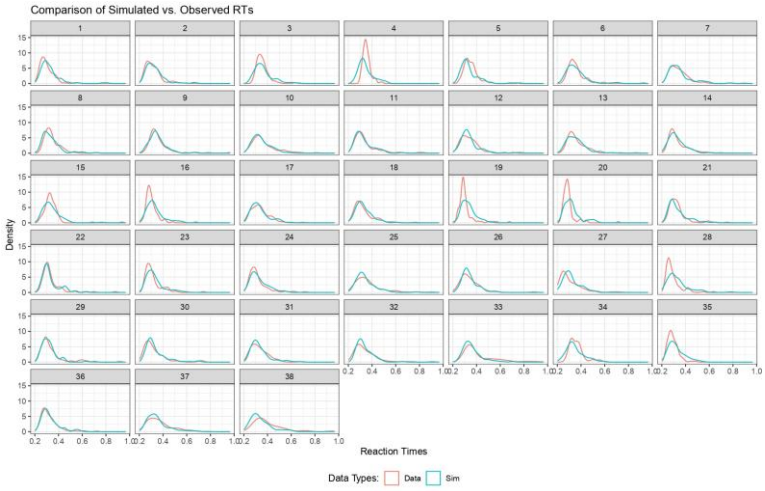
| Measure | Fixed Effect | Estimate | SE | df | t | p | 95% CI |
|---|---|---|---|---|---|---|---|
| Simple Reaction Time | Intercept | 325.71 | 11.16 | 28.58 | 29.18 | <.001*** | [302.87, 348.55] |
| | State (Hungover vs. Sober) | 30.72 | 14.41 | 28.58 | 2.13 | .042* | [1.23, 60.20] |
| | Session (2 vs. 1) | 34.21 | 14.62 | 29.13 | 2.34 | .026* | [4.32, 64.10] |
| | State by Session | -66.72 | 24.84 | 17.31 | -2.69 | .015* | [-119.06, -14.39] |

**Drift Diffusion Modelling of Decision Parameters.** HDDM was used to fit the observed RT data. Posterior predictive checks confirmed that the HDDM captured group- and individual-level RT distributions effectively, as shown in Figure 6. The high degree of alignment between observed and simulated data suggested that the estimated cognitive parameters were reliable.

**Figure 6**

*Posterior Predictive Checks for HDDM*



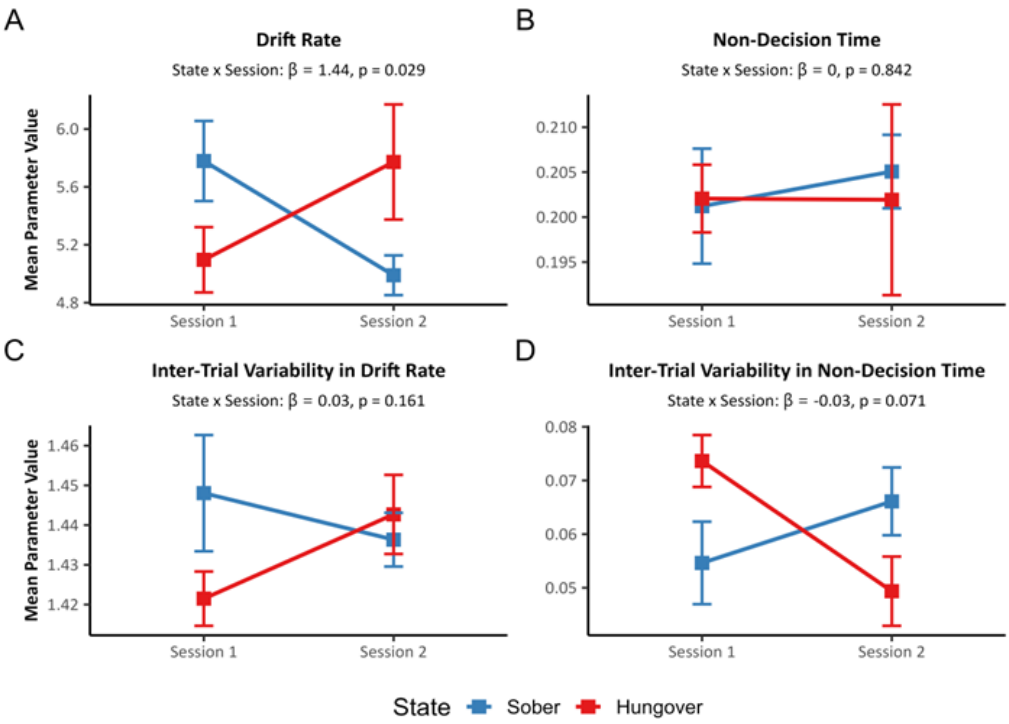



*Note.* Simulated and observed RT distributions are overlaid to validate model fit. Panel A compares the reaction time (RT) distributions between observed data (red line) and simulated data (blue line) for each of the 19 individual participants across both sessions. Panel B shows the aggregated comparison across all participants.

An LMM was fitted for each of the four primary DDM parameters($v, t_0, sv, st_0$) to examine the effects of state, session, and their interaction. The interaction between the main effects of state and session is illustrated in Figure 7, and a summary of the fixed effects is presented in Table 3.

**Figure 7**

*Interaction plots for DDM parameter estimates.*



*Note*. Panels show (A) drift rate, (B) non-decision time, (C) inter-trial variability in drift rate, and (D) inter-trial variability in non-decision time. The mean parameter value for each state across both sessions is plotted with error bars depicting the standard error of the mean. Intersecting lines indicate a state-by-session interaction effect, and the corresponding $\beta$ coefficient and $p$-value are displayed in each subtitle.

**Table 3**

*Fixed Effects Parameter Estimates for DDM Parameters by State and Session*

| Parameter | Fixed Effect | Estimate | SE | df | p | t | 95% CI |
|---|---|---|---|---|---|---|---|
| $v$ | Intercept | 5.78 | 0.27 | 28.22 | <.001*** | 21.31 | [5.22, 6.33] |
| | State | -0.68 | 0.35 | 28.22 | .061 | -1.95 | [-1.40, 0.03] |
| | Session (2 vs. 1) | -0.73 | 0.36 | 28.79 | .048* | -2.06 | [-1.46, -0.01] |
| | State by Session | 1.44 | 0.60 | 16.79 | .029* | 2.39 | [0.17, 2.72] |
| $t_0$ | Intercept | 0.20 | 0.01 | 21.61 | <.001*** | 32.60 | [0.19, 0.21] |
| | State | 0.00 | 0.01 | 21.61 | .917 | 0.11 | [-0.02, 0.02] |
| | Session (2 vs. 1) | 0.00 | 0.01 | 22.02 | .645 | 0.47 | [-0.01, 0.02] |
| | State by Session | -0.00 | 0.02 | 18.32 | .842 | -0.20 | [-0.04, 0.03] |
| $sv$ | Intercept | 1.45 | 0.01 | 27.82 | <.001*** | 145.28 | [1.43, 1.47] |
| | State | -0.03 | 0.01 | 27.82 | .049* | -2.06 | [-0.05, -0.00] |
| | Session (2 vs. 1) | -0.01 | 0.01 | 28.38 | .406 | -0.84 | [-0.04, 0.02] |
| | State by Session | 0.03 | 0.02 | 18.46 | .161 | 1.46 | [-0.01, 0.08] |
| $st_0$ | Intercept | 0.05 | 0.01 | 24.55 | <.001*** | 7.87 | [0.04, 0.07] |
| | State | 0.02 | 0.01 | 24.55 | .044* | 2.12 | [0.00, 0.04] |
| | Session (2 vs. 1) | 0.01 | 0.01 | 25.22 | .272 | 1.12 | [-0.01, 0.03] |
| | State by Session | -0.03 | 0.02 | 16.45 | .071 | -1.93 | [-0.07, 0.00] |

For drift rate *(v)*, the model revealed a significant state by session interaction, $t(16.79) = 2.39$, $p = .029$. The effect of the state on $v$ was not significantly different between the two studies, and the significant interaction effect made it difficult to interpret the significant main effect of session, $t(28.79) = -2.06$, $p = .048$. For non-

decision time (**$t_0$**), reflecting perceptual and motor processes, no significant main effects or interactions were found (all $p > 0.05$).

The state-by-session interaction was insignificant for both inter-trial variability in drift rate (*sv*; $p = .161$) and non-decision time ($st_0$; $p = .071$). Contrarily, a significant main effect of state was observed in *sv* ($t(27.82) = -2.06$, $p = .049$), with lower values reported in the hangover state, and $st_0$ ($t(24.55) = 2.12$, $p = .044$), with higher values reported in the hangover state.

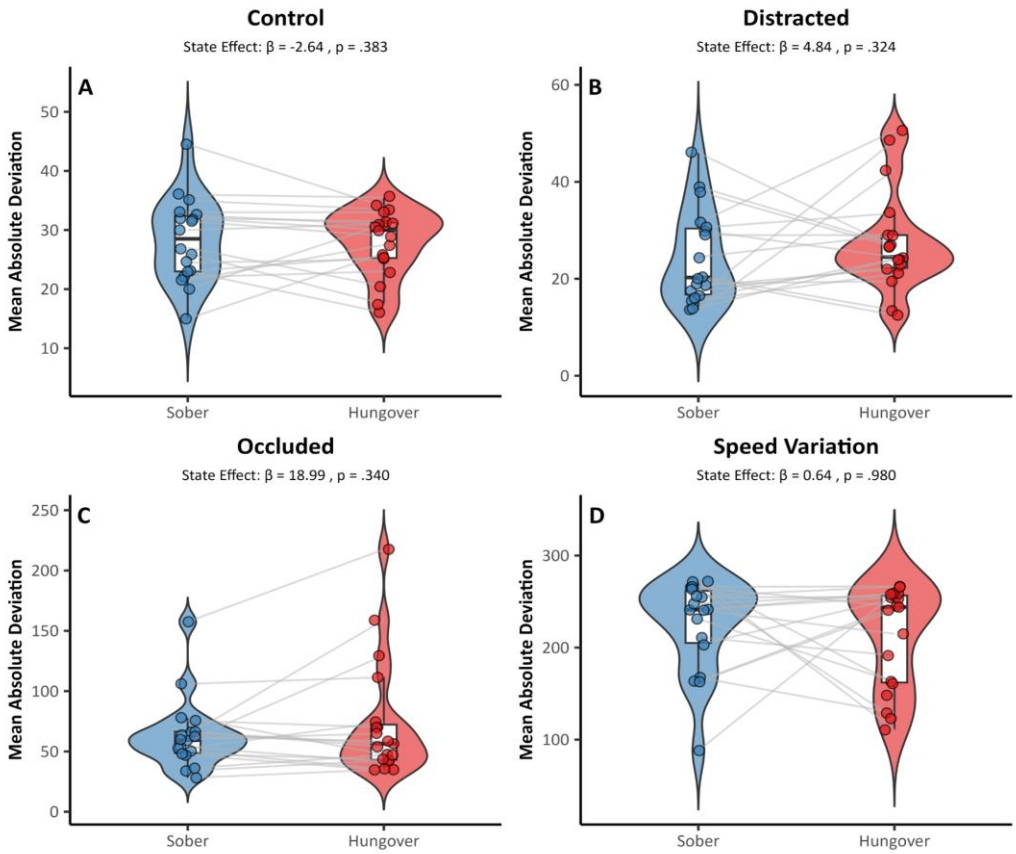
**Temporal Estimation Performance**

We next assessed the effect of hangover on motor timing using an anticipatory timing task. LMMs were fitted to the data using **t**he MAD as a measure of estimation variability as reported in Table 4. Contrary to our hypothesis, no significant main effect of state emerged for temporal estimation variability in any of the four task conditions: Control ($\beta = -2.64$, $p = .383$), Distracted ($\beta = 4.84$, $p = .324$), Occluded ($\beta = 18.99$, $p = .340$), and Speed Variation ($\beta = 0.64$, $p = .980$)). Session and state by session interaction effects were also non-significant across all conditions ($p > 0.05$).

Individual-level data showed considerable variability as shown in **Figure 8,** and while the direction of the effect in some conditions (e.g., Occluded) was consistent with our hypothesis, the large range of confidence intervals (95 % CI = [-21.35,59.33]) made it hard to reliably report any effects.

**Figure 8**

*Variability in temporal estimation error across conditions*




*Note.* Violin plots depict MAD in all conditions of the motor timing task. Box plots represent the mean and 95 % confidence intervals. Points represent individual participant data, connected by lines across states. Subtitles display the LMM parameter estimates ($\beta$) and $p$-values for the main effect of state.

## Table 4

*Linear Mixed-Effects Model Results for Temporal Estimation Variability*

| Self-Rated Measure | Fixed Effect | Estimate | SE | df | t | p | 95% CI |
|---|---|---|---|---|---|---|---|
| Control | Intercept | 31.14 | 2.36 | 23.28 | 13.19 | <.001*** | [26.26, 36.02] |
| | State | -2.64 | 2.97 | 23.28 | -0.89 | .383 | [-8.78, 3.50] |
| | Session (2 vs. 1) | -4.61 | 3.00 | 23.80 | -1.54 | .137 | [-10.80, 1.58] |
| | State by Session | 3.11 | 5.45 | 17.10 | 0.57 | .575 | [-8.37, 14.59] |
| Distracted | Intercept | 22.29 | 3.84 | 32.69 | 5.80 | <.001*** | [14.47, 30.10] |
| | State | 4.84 | 4.83 | 32.69 | 1.00 | .324 | [-5.00, 14.68] |
| | Session (2 vs. 1) | 3.45 | 4.91 | 32.73 | 0.70 | .488 | [-6.55, 13.45] |
| | State by Session | -2.86 | 7.23 | 16.98 | -0.40 | .697 | [-18.11, 12.39] |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Occluded | Intercept | 54.93 | 15.51 | 23.27 | 3.54 | .002** | [22.87, 86.99] |
| | State | 18.99 | 19.51 | 23.27 | 0.97 | .340 | [-21.35, 59.33] |
| | Session (2 vs. 1) | 13.88 | 19.69 | 23.78 | 0.70 | .488 | [-26.78, 54.54] |
| | State by Session | -15.51 | 35.78 | 17.12 | -0.43 | .670 | [-90.96, 59.94] |
| Speed Variation | Intercept | 222.57 | 20.20 | 33.00 | 11.02 | <.001*** | [181.47, 263.68] |
| | State | 0.64 | 25.42 | 33.00 | 0.03 | .980 | [-51.09, 52.36] |
| | Session (2 vs. 1) | 3.66 | 25.85 | 33.00 | 0.14 | .888 | [-48.93, 56.24] |
| | State by Session | -32.22 | 36.25 | 33.00 | -0.89 | .381 | [-105.98, 41.54] |

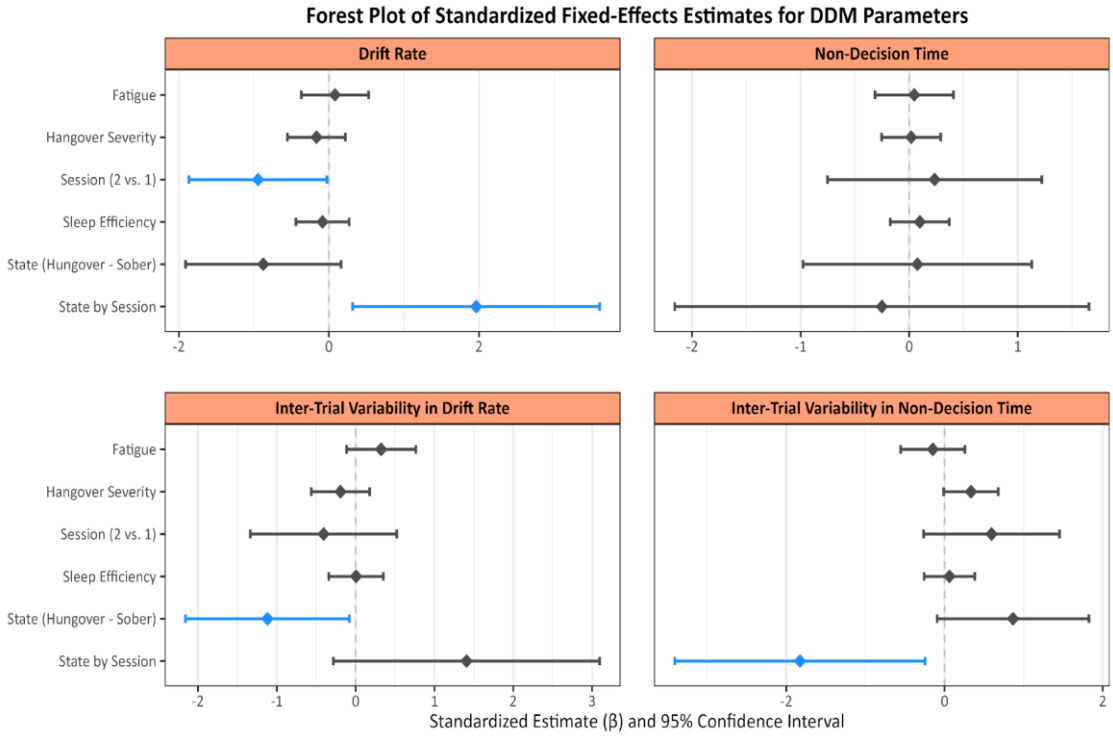
**Exploratory Covariate Analysis**

Finally, to have a better understanding of the factors mediating the observed effects, the LMMs were extended to include hangover severity, sleep efficiency, and fatigue as predictors, with all continuous variables undergoing a *z*-score transformation. This enabled a direct comparative analysis of each predictor's effect sizes.

Figure 9 presents the standardised regression results for the DDM parameters. After including all the covariates, only the state by session interaction remained significant for both drift rate($v$) ($t(16.67) = 2.52$, $p = .022$) and inter-trial variability in non-decision time ($st_0$) ($t(14.49) = -2.47$, $p = .027$). Surprisingly, the main effect of state also emerged to be significant for inter-trial variability in drift-rate ($sv$) ($t(28.81) = -2.20$, $p = .036$), indicating lower values in the hungover state. However, none of the self-reported covariates (Hangover severity, fatigue, sleep Efficiency) emerged as significant predictors of any DDM parameters (all $p > 0.05$), suggesting that observed effects in the decision-making process are not explained by the self-reported measures.

**Figure 9**.

*Forest plot of standardised regression coefficients (β) from the LMMs for DDM parameters.*

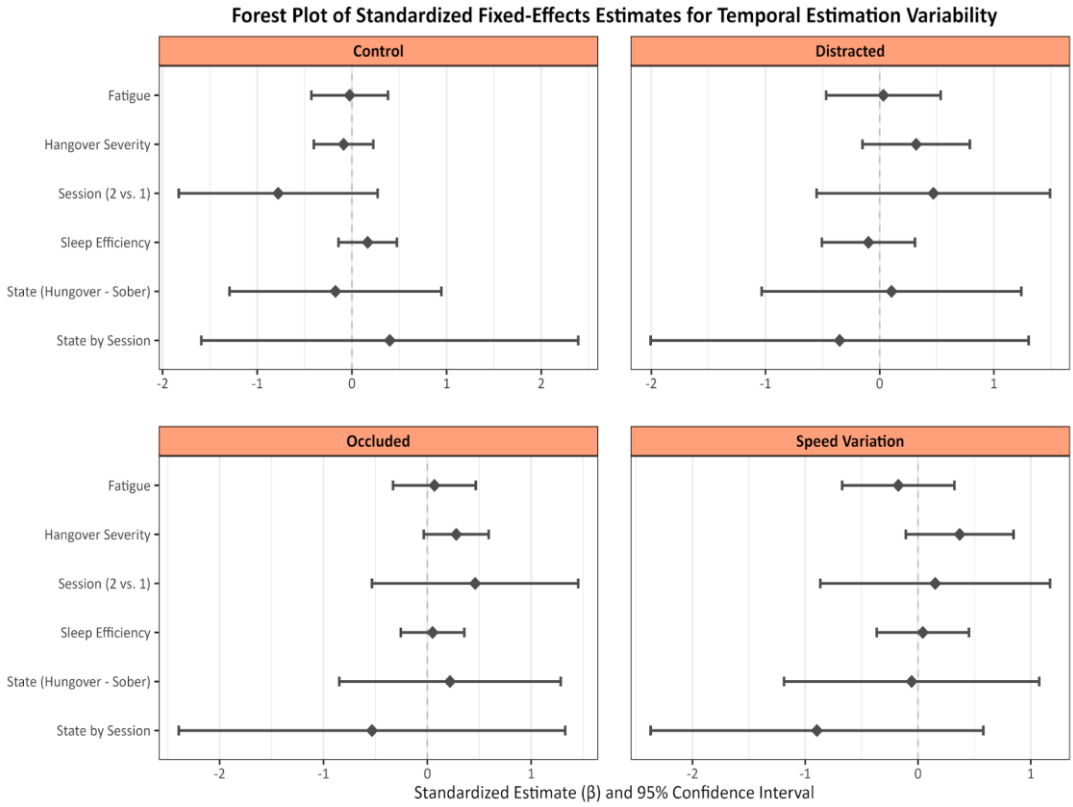



*Note.* Each panel displays the results for a different DDM parameter. Points represent the estimated effect of each predictor, and horizontal lines indicate the 95% confidence intervals. Effects are considered significant ($p < .05$) if their confidence interval does not cross the zero line and are highlighted in blue.

For the MAD measures in the anticipatory timing task, a similar pattern was observed in Figure 10**.** Once again, the three covariates emerged as non-significant predictors of the temporal estimation variability across the four conditions. Although small changes in estimates' direction were noted, the overlap of their confidence intervals with zero favoured the null hypothesis. The full statistical output for these models can be found in Supplementary Tables 1 and 2.

**Figure 10**

*Forest plot of Standardised regression coefficients (β) from the LMMs for temporal estimation variability*




*Note.* Each panel displays the results for a different task condition. Points represent the estimated effect of each predictor, and horizontal lines indicate the 95% confidence intervals. Effects are considered significant ($p < .05$) if their confidence interval does not cross the zero line and are highlighted in blue.

# Discussion

This study aimed to investigate the effect of hangover-related impairment on cognitive functions relevant to driving, such as psychomotor speed, decision-making and temporal estimation. Studies have adopted different approaches to validate the effects of alcohol related hangover on these domains, including subjective and objective scales, with recent research focusing on decision-making modelling, and found varied results (Verster et al., 2014; Ayre et al., 2021; Grange et al., 2016; Stock et al., 2017). To address identified gaps in the literature, we had hypothesised that participants impaired by the effects of alcohol hangover would show changes in psychomotor speed, evidence accumulation and variability in temporal estimation, with sleep and fatigue potentially mediating these effects.

**SRTT Findings**

We observed a significant increase in the reaction time(RT) in the hangover condition ($\beta = 30.72$, $p = .042$). The direction of the main effect was consistent with previous research across various task paradigms and sample sizes, suggesting a slowing of psychomotor speed as a marker for hangover-related impairment (Grange et al., 2016; Van Ravenzwaaij et al., 2012). Devenney et al.(2019) reported that individuals in a hangover state exhibited increased reaction times compared to a control group in a different choice reaction time task. For driving safety, this 31 ms response delay might mean the difference between avoiding/causing a collision, potentially explaining epidemiological links between hangover and accident risk (Verster et al., 2014). However, there was also a significant state-by-session interaction effect, which made it difficult to interpret the main effects. While the session effect has not been explored extensively, performance in cognitive tasks is potentially influenced by learning effects, often manifesting as improved speed or accuracy with repeated exposure to the task paradigms. In addition, practice leads to distinct behavioural changes manifested by improved skill level and reduced cognitive load (Boksem et al., 2005; Dutilh et al., 2009). In an analysis by Murgia et al. (2020), the increase in reaction time during hangover was reversible when the same subjects were re-tested in a non-hangover state (a within-subjects comparison). Participants may also compensate for their hangover state by becoming more cautious in their responses, leading to improved performance in subsequent sessions (Grange et al., 2016).

**DDM Analysis**

This study is one of the few studies in the literature to apply DDM to the hangover state, although this model has previously been used to study acute alcohol intoxication. We employed a hierarchical Bayesian Ratcliff diffusion model in our study implemented in Stan which offered a robust and flexible approach to analyse our behavioural data. It offers several critical advantages over simpler DDM fitting techniques like the EZ-DDM (van Ravenzwaaij et al., 2017) by simultaneously estimating individual decision measures and the posterior group distributions from which they are drawn. Hierarchical models have been replicated well in other modelling studies and are known for their reduced parameter variability (Stafford et al., 2020). The major drawback of this approach is its computational complexity. Implementation of Bayesian inferences further enhances the validity of our group estimates since it supports non-central estimations, non-linear posterior distributions and better model diagnostics.

Our study found a significant crossover effect between state and session for drift rate ($v$) ( $p$ = .029), indicating that hangover effects on evidence accumulation may be mediated by session order. Contrary to our hypothesis, hangover state or session had no significant main effect on non-decision time ($t_0$). These findings suggest that peripheral processes like motor encoding and execution were reasonably stable across states and sessions.

In the present study, we examined not only the primary diffusion model parameters ($v$, $t_0$) but also the corresponding inter-trial variability components ($sv$, $st_0$). The inclusion of these parameters was motivated theoretically by prior studies that suggested assessing them as meaningful measures of fluctuations in cognitive states due to attentional lapses, perceptual noise and shifts in task motivation (Ratcliff & Rouder, 1998; Ratcliff & Tuerlinckx, 2002).

Specifically, $sv$, the inter-trial variability in drift-rate, helps capture fluctuations in the rate of evidence accumulation across trials (McVay & Kane, 2012; Hawkins et al., 2015). Modelling research suggests improved parameter recovery and interpretability when $sv$ is included (Lerche et al., 2016; Ratcliff & Tuerlinckx, 2002; Steingroever et al., 2020). Similarly, $st_0$, the variability in non-decision time estimate, accounts for variations in motor encoding processes, which are also susceptible to hangover-related cognitive impairment (Sandry & Ricker, 2022; Nguyen & Reinagel, 2022).

As anticipated, the trial-to-trial variability parameters revealed a completely different story. The state effect was robust in predicting Inter-trial variability in drift rate (*sv*), with significantly lower values estimated in the hangover state, suggesting a more conservative accumulation process, possibly due to reduced cognitive flexibility or adaptability in the hangover state. Inter-trial variability in non-decision time ($st_0$) was also affected with significantly higher estimates in the hangover state. This could indicate greater inconsistency in perceptual or motor execution across trials. The opposing nature of these effects could reflect differential sensitivity of central versus peripheral processes to hangover (Stock et al., 2017).

The decision to include both of them also aligned with recent implementations of the Ratcliff diffusion model that advocate for their inclusion. The potential to capture underlying perceptual noise and assess true cognitive effects enhances the sensitivity and reproducibility of our model and sets it apart from other implementations such as EZ-DDM and Robust-EZ, which only estimate mean-level parameters and suffer from poor parameter recovery, particularly for non-decision components (Wagenmakers et al., 2007; Grange et al., 2016; Stafford et al., 2020).

While the aggregated model distributions closely represented observed data, individual posterior distributions had considerable inter-participant heterogeneity, limiting the scope of our findings. Table 5 contextualises our findings with respect to previous hangover research with DDM models.

**Table 5**

*Comparison of DDM results across studies*

| DDM Parameter | Grange et al. (2016) | Stock et al. (2017) | Present Study (N =19) |
|---|---|---|---|
| Drift Rate ($v$) | Reduced<br>$p < .001$, $d = -0.76$ | Increased<br>$p < .025$, $d = 0.66$ | Non-significant<br>$p = .061$, $d = 0.02$ |
| Boundary Separation (a) | Increased<br>$p = .005$, $d = 0.56$ | No significant effects | Fixed at 1.0 |
| Non-Decision Time ($t_0$) | Marginal<br>$p = .053$, $d = -0.38$ | Inconclusive<br>$p = .08$, d=0.42 | Non-significant |

| | | | |
|---|---|---|---|
| Variability in Drift Rate (*sv*) | Not reported | Not reported | $p$ = .036, $d$ = -0.40 |
| Variability in Non-Decision Time ($st_0$) | Not reported | Not reported | $p$ = .027, $d$ = 0.35 |

*Note.* Other studies reported *p*-values from paired t-tests, while we reported *p*-values from the regression analysis. Cohen's *d* was used to estimate the effect size.

**Alcohol Hangover vs Alcohol Intoxication**

Acute alcohol intoxication studies have also inconsistently reported dose-dependent DDM disruptions. Drift rate has been observed to be altered in alcohol intoxication, while the hangover state may preserve it. Van Ravenzwaaij et al. (2012) reported reduced drift rates at 0.06% BAC during a perceptual decision task and also found prolonged non-decision time (suggesting slower sensory/motor processing) but unchanged boundary separation. Stock et al. (2017) surprisingly found that a hangover state increased the drift rate while acute intoxication tended to reduce it. Alcohol may modulate brain neurotransmitters in opposite directions depending on dose and state. Alcohol is known to affect brain chemistry by interacting with two key neurotransmitters, namely GABA and glutamate. Inhibitory effects of GABA, which lead to sedation, are intensified while glutamate's excitatory effects are inhibited. This interplay may be variably affected in intoxication and hangover states. Hangover's milder, delayed effects may preferentially target motor output rather than decision circuits (Stock et al., 2017; Polli et al., 2024; Dry et al., 2012).

Notably in this study, prolongation of RT tended to happen despite limited impact on DDM parameters, suggesting selective influence of hangover on psychomotor execution rather than central decision processes. This discrepancy parallels observations in neurological diseases where basal ganglia dysfunction slows RTs without altering evidence accumulation (Herz et al., 2016).

**Motor Timing Task Findings**

Since the SRTT primarily assesses basic psychomotor speed, it lacks the complexity needed to evaluate motor control. To address this shortcoming, we used the visuospatial timing task to help evaluate variability in temporal estimation. The measure of choice here was the MAD in temporal estimation error. This measure was chosen because of its capacity to capture trial-by-trial fluctuations in performance, making it

suitable for detecting within-subject variability (Rodrigues et al., 2009; Stolz et al., 2023)

The study failed to find any significant hangover-related effects in temporal estimation variability in the anticipatory timing task, depicting the relative resilience of internal timing models. These models are understood to be encoded within cortico-striatal and cerebellar networks and continue to remain functionally intact under acute disruptions of state (Buhusi & Meck, 2005; Matell & Meck, 2004). This interpretation aligns with other theories proposing that once temporal priors are established through experience, their application is mainly instinctive and less reliant on attentional control (Herz et al., 2016; Menceloglu et al., 2021). Evidence suggests that domains that rely on stimulus processing, active control, and adaptation are more susceptible to disruption by hangover state (Verster et al., 2003; Gunn et al., 2018). A surprising finding was that there were no significant effects of the hangover state on the MAD across all task conditions, including those with increased cognitive load, such as occlusion and speed variation. More complex tasks involving dynamic estimation demands may be more effective in uncovering subtle differences in motor timing. These findings are somewhat in line with prior studies that show tasks involving estimation of motor timing are less susceptible to minor differences in state (Buhusi & Meck, 2005; Matell & Meck, 2004). The lack of association between subjective symptom severity and MAD in our study helps underline the challenges of linking self-perceived impairment with task-based outcomes.

**Limitations**

The study employed a semi-naturalistic, within-subjects, crossover design. Such study designs have previously been used in hangover research because of their ecological validity and high degree of experimental control. However, several limitations exist. Firstly, the small sample size and mild hangover severity limited our ability to detect cognitive deficits observed in more severe hangover conditions. The semi-naturalistic design also introduces variability in alcohol type and doses, drug usage, drinking duration and sleep patterns. These factors may be difficult to control and can possibly confound inferences on cognitive effects associated with hangovers. There is also an expectancy effect since participants are already aware they are going to be assessed, which might influence the findings. The skewed gender ratio in favour of females made it hard to investigate potential gender-based differences in symptom reporting and task performance.

Moreover, the HDDM operates under the assumption that evidence accumulation is independent and sequential, which might not fully capture the complexity of real-world cognition, especially in states like hangover. Participants may also adopt certain compensatory strategies to maintain performance, masking deficits in the decision-making process. These limitations motivated us to combine HDDM with mixed-effects regression models to help address some of these limitations.

The testing battery that we used also had a limited scope. While a simple task like the SRTT offers sensitivity and specificity towards the primary outcome measure, it also limits the range of cognitive domains it can assess. The motor timing task provided higher task complexity and the ability to measure performance under additional cognitive load (e.g., occlude and distract conditions) but had limited generalisability to real-world driving when compared with simulator studies. Executive functions such as working memory and response inhibition could also not be assessed.

**Subjective vs Objective Performance**

The study did not find any concrete findings to support our hypothesis that subjective ratings of fatigue, sleep and hangover severity would be significant predictors of cognitive and motor task performance, even after accounting for the main effect of hungover state. Single symptom severity and overall hangover score (AHSS) were used as covariates in our analysis (Penning et al., 2012b, 2013). Most of our subjects exhibited mild hangovers. It is possible to under- or overestimate aggregate symptom scores, which may lead to deviation from the true hangover effect, leading to a preference for single-item measurements. We used sleep efficiency instead of sleep duration, as it has been found to be a better indicator of overall psychological and physical health (Kohyama et al. 2021). We tried to explore the relation using linear regression modelling for DDM parameters

Through varied designs and methodological differences, previous studies have explored the independent and covariate effects of sleep and fatigue on cognitive performance in hangover. Negative association between AHS and poor sleep has been reported previously, in addition to cognitive decline (Gunn et al. 2018; Ayre et al. 2021; Devenney et al. 2019).

DDM modelling has been previously used to assess effects of sleep deprivation on decline in psychomotor vigilance (Ratcliff et al., 2009, 2011), but the complex interplay of hangover, fatigue and sleep deprivation simultaneously had not been examined using this modelling approach. Sleep and fatigue are known confounding

factors while assessing cognitive performance, and by including these factors as covariates, we can help distinguish the contributions of different yet overlapping factors. The findings of this study are in contrast to previous studies, which have reported a reduction in drift parameters with sleep deprivation. A few factors may explain the lack of interaction effect in our study. Firstly, we have used a different modelling approach and task paradigms. Subjective ratings of sleep deprivation may be lower in our study, and finally, sleep deprivation may have non-homogeneous circadian elements. It has also been reported that the drift rate may reduce for easier tasks but remain unaffected for difficult ones (Menz et al, 2012).

**Implications**

The present study was an extension of previous Hangover research. To our understanding, no other study has combined HDDM with mixed effects regression analysis to assess temporal estimation variability in hangover state. While the effects of hangover itself were limited to certain domains rather than affecting all aspects of driving-related cognition, our research presents a novel way of examining these diverse effects. Even though we could not establish a definitive link between subjective factors like hangover severity, fatigue and sleep, the disconnect between these measures and task performance warrants further examination. It is possible that findings from cognitive tasks with a learning component may be influenced by session or order-related effects and are less sensitive to disruptions in state.

**Future Research**

The methodological design developed in this thesis leaves scope for further exploring an ideal paradigm that investigates these cognitive deficits. DDM model implementations can potentially be improved by increasing the number of parameters and more complex designs involving more iterations and sampling. Research can be done in a broader set of populations with representative samples from various age groups, sex and education levels. Investigation of individual symptom severity across hangover scales may require a more sophisticated neurophysiological and imaging approach. Various mobile apps and wearable devices can be developed to monitor the sleep and hangover effects, and these can be used to develop strategies for improving session timings. Modelling approaches can be combined with a model-free approach to real-world scenarios to capture the impairments. Longitudinal studies, especially those combining neuroimaging with DDM, need to be developed.

**Conclusion**

The study found some basic changes in psychomotor speed and slower evidence accumulation, but the main effects could have been impaired by the interaction of session and state. Despite these crossover interactions, this study adds value to the limited literature as it provides a deeper insight and novel analytical approach to the effect of hangover on impairment of simple attentional and motor processes(reaction time) and higher-order cognitive control mechanisms involved in inhibitory tasks, as shown by the HDDM model and anticipatory timing task. It is thus imperative that individual subjective components like sleep and fatigue are studied separately rather than having a generalised view of overall performance in a hangover state. Even small RT delays could have a significant impact on driving ability, but further research is needed to confirm whether these effects are dependent on hangover severity or task complexity.

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# Appendix A

## Consensus Sleep Diary

What time did you get into bed last night?

What time did you try to go to sleep last night (e.g., 11:55 pm or 23:55)? (Record the time that you began "trying" to fall asleep.)

How long did it take to fall asleep (minutes, e.g., 30)? (Beginning at the time you wrote in question 2)

How many times did you wake up, not counting your final awakening? (How many times did you wake up between the time you first fell asleep and your final awakening?)

- None
- 1
- 2
- 3
- 4
- 5
- 6 or more

In total, how long did these awakenings last? (What was the total time you were awake between the time you first fell asleep and your final awakening. For example, if you woke 3 times for 20 minutes, 35 minutes, and 15 minutes, add them all up (20+35+15 = 70))

What time was your final awakening (e.g., 7:21 am)? (Record the last time you woke up in the morning.)

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What time did you get out of bed for the day (e.g., 8:50 am)? (What time did you get out of bed with no further attempt at sleeping? This may be different from your final awakening time, e.g. you may have woken up at 6:35 a.m. but did not get out of bed to start your day until 7:20 a.m.)

How would you rate the quality of your sleep last night?

Very poor — Very good

0 10 20 30 40 50 60 70 80 90 100

# Appendix B

## Alcohol Hangover Symptom Severity Questionnaire

Please move the slider accordingly to indicate how you feel today for each of the following symptoms:

Absent — Extreme

0 10 20 30 40 50 60 70 80 90 100

Fatigue (being tired)

Apathy (lack of interest)

Concentration problems

Clumsiness

Confusion

Thirst

Sweating

Shivering

Stomach pain

Nausea

Dizziness

Heart Pounding

Headache

Sensitivity to light

Anxiety

# Appendix C

**Supplementary Table 1**

*Fixed Effects Parameter Estimates for Temporal Estimation Variability*

| Task Condition | Fixed Effect | Estimate | df | t | p | 95% CI |
|---|---|---|---|---|---|---|
| Control | Intercept | 20.11 | 19.83 | 1.87 | .077 | [-2.38, 42.60] |
| | State (Hungover - Sober) | -1.09 | 24.69 | -0.32 | .751 | [-8.09, 5.91] |
| | Session (2 vs. 1) | -4.88 | 21.60 | -1.54 | .138 | [-11.45, 1.70] |
| | Sleep Efficiency | 12.84 | 18.34 | 1.13 | .273 | [-11.02, 36.69] |
| | Fatigue | -0.01 | 21.87 | -0.12 | .905 | [-0.11, 0.10] |
| | Hangover Severity | -0.05 | 15.19 | -0.60 | .559 | [-0.21, 0.12] |
| | State by Session | 2.50 | 16.52 | 0.42 | .677 | [-9.96, 14.96] |
| Distracted | Intercept | 31.58 | 29.77 | 1.38 | .177 | [-15.13, 78.28] |
| | State (Hungover - Sober) | 1.03 | 29.36 | 0.19 | .853 | [-10.25, 12.31] |
| | Session (2 vs. 1) | 4.66 | 29.46 | 0.94 | .355 | [-5.48, 14.81] |
| | Sleep Efficiency | -12.09 | 29.69 | -0.49 | .625 | [-62.11, 37.94] |

| | | | | | | |
|---|---|---|---|---|---|---|
| | Fatigue | 0.01 | 29.75 | 0.13 | .896 | [-0.19, 0.22] |
| | Hangover Severity | 0.27 | 20.55 | 1.41 | .172 | [-0.13, 0.66] |
| | State by Session | -3.48 | 13.07 | -0.46 | .655 | [-19.90, 12.95] |
| Occluded | Intercept | 25.95 | 21.50 | 0.38 | .710 | [-116.93, 168.83] |
| | State (Hungover - Sober) | 8.76 | 25.87 | 0.42 | .677 | [-34.03, 51.56] |
| | Session (2 vs. 1) | 18.48 | 23.12 | 0.96 | .348 | [-21.40, 58.36] |
| | Sleep Efficiency | 25.02 | 20.25 | 0.34 | .735 | [-126.70, 176.73] |
| | Fatigue | 0.11 | 23.83 | 0.35 | .726 | [-0.54, 0.77] |
| | Hangover Severity | 0.95 | 16.70 | 1.88 | .077 | [-0.12, 2.01] |
| | State by Session | -21.40 | 17.58 | -0.60 | .554 | [-96.12, 53.32] |
| Speed Variation | Intercept | 201.63 | 30.00 | 1.66 | .107 | [-45.87, 449.12] |
| | State (Hungover - Sober) | -2.97 | 30.00 | -0.10 | .919 | [-62.38, 56.45] |

| | | | | | | |
|---|---|---|---|---|---|---|
| | Session (2 vs. 1) | 8.00 | 30.00 | 0.31 | .762 | [-45.51, 61.51] |
| | Sleep Efficiency | 27.63 | 30.00 | 0.21 | .833 | [-237.58, 292.84] |
| | Fatigue | -0.38 | 30.00 | -0.72 | .479 | [-1.45, 0.70] |
| | Hangover Severity | 1.64 | 30.00 | 1.58 | .125 | [-0.48, 3.76] |
| | State by Session | -47.08 | 30.00 | -1.24 | .224 | [-124.55, 30.40] |

**Supplementary Table 2**

*Fixed Effects Parameter Estimates for DDM Parameters*

| DDM Parameter | Fixed Effect | Estimate | df | t | p | 95% CI |
|---|---|---|---|---|---|---|
| Drift Rate | Intercept | 6.52 | 26.11 | 3.99 | <.001*** | [3.16, 9.88] |
| | State (Hungover - Sober) | -0.71 | 29.19 | -1.72 | .095 | [-1.56, 0.13] |
| | Session (2 vs. 1) | -0.77 | 27.24 | -2.11 | .045* | [-1.52, -0.02] |
| | Sleep Efficiency | -0.86 | 25.54 | -0.49 | .626 | [-4.44, 2.72] |

| | | | | | | |
|---|---|---|---|---|---|---|
| | Fatigue | 0.00 | 29.37 | 0.37 | .715 | [-0.01, 0.02] |
| | Hangover Severity | -0.01 | 18.07 | -0.90 | .382 | [-0.04, 0.02] |
| | State by Session | 1.60 | 16.67 | 2.52 | .022* | [0.26, 2.94] |
| Non-Decision Time | Intercept | 0.18 | 19.16 | 6.98 | <.001*** | [0.13, 0.24] |
| | State (Hungover - Sober) | 0.00 | 25.56 | 0.15 | .883 | [-0.02, 0.02] |
| | Session (2 vs. 1) | 0.00 | 21.88 | 0.49 | .627 | [-0.01, 0.02] |
| | Sleep Efficiency | 0.02 | 17.53 | 0.76 | .457 | [-0.04, 0.08] |
| | Fatigue | 0.00 | 20.34 | 0.27 | .789 | [-0.00, 0.00] |
| | Hangover Severity | 0.00 | 15.12 | 0.15 | .886 | [-0.00, 0.00] |
| | State by Session | -0.00 | 18.41 | -0.28 | .785 | [-0.04, 0.03] |
| Inter-Trial Variability in Drift Rate | Intercept | 1.44 | 24.68 | 25.38 | <.001*** | [1.32, 1.55] |

| | | | | | | |
|---|---|---|---|---|---|---|
| | State (Hungover - Sober) | -0.03 | 28.81 | -2.20 | .036* | [-0.06, -0.00] |
| | Session (2 vs. 1) | -0.01 | 26.29 | -0.90 | .377 | [-0.04, 0.02] |
| | Sleep Efficiency | 0.00 | 23.98 | 0.04 | .971 | [-0.12, 0.13] |
| | Fatigue | 0.00 | 28.16 | 1.51 | .142 | [-0.00, 0.00] |
| | Hangover Severity | -0.00 | 17.54 | -1.09 | .290 | [-0.00, 0.00] |
| | State by Session | 0.04 | 17.24 | 1.76 | .097 | [-0.01, 0.09] |
| Inter-Trial Variability in Non-Decision Time | Intercept | 0.04 | 22.56 | 1.12 | .277 | [-0.04, 0.12] |
| | State (Hungover - Sober) | 0.02 | 27.91 | 1.85 | .075 | [-0.00, 0.04] |
| | Session (2 vs. 1) | 0.01 | 24.58 | 1.43 | .166 | [-0.01, 0.03] |
| | Sleep Efficiency | 0.02 | 21.71 | 0.41 | .687 | [-0.07, 0.10] |

| | | | | | | |
|---|---|---|---|---|---|---|
| | Fatigue | -0.00 | 26.95 | -0.75 | .459 | [-0.00, 0.00] |
| | Hangover Severity | 0.00 | 14.72 | 2.07 | .056 | [-0.00, 0.00] |
| | State by Session | -0.04 | 14.49 | -2.47 | .027* | [-0.07, -0.01] |